

Homework 2 - Creative mathematical worlds

Five questions. Each question is worth 10 points (50 total).

Write a clear argument when requested. Valid alternative proofs and clear detours are welcome. Follow each exercise's specific requirements. This handout contains no solutions.

Question 1. The two-button robot

A robot's integer display starts at 0. Button P adds 8 and button Q subtracts 6. You may press either button any nonnegative finite number of times, in any order, or press no buttons.

Characterize exactly which integers can appear. Prove that every claimed value is reachable and that no other value is reachable.

Question 2. One key for every visitor?

There are three visitors $V=\{\text{Ada}, \text{Bo}, \text{Cy}\}$ and three keys $K=\{\text{red}, \text{green}, \text{blue}\}$. The predicate $R(v,k)$ means that key k opens visitor v 's locker.

Write quantified statements for (a) every visitor has at least one working key and (b) there is a single key that works for every visitor. Does (a) imply (b)? Give a fully specified counterexample or a proof.

Question 3. Three self-referential cards

Three cards carry the following statements. Card A: 'Card B is false.' Card B: 'Card C is false.' Card C: 'Card A and Card B are both false.'

Assign true or false to each card so that each assigned value agrees with what the card says. Prove that your assignment is the only consistent one.

Question 4. The archive's invert-selection button

An archive has a finite set U of files. Clicking 'Invert' changes a selected subset $A \subseteq U$ to $T(A) = U \setminus A$.

Prove that $T(T(A)) = A$ for every subset A . If $U \neq \emptyset$, can any selection satisfy $T(A) = A$? Prove your answer, and explain separately what happens when $U = \emptyset$.

Question 5. The paired shelf labels

A librarian chooses 13 distinct shelf labels from $\{1, 2, \dots, 24\}$. Prove that two chosen labels differ by exactly 12.

If you use a counting or pigeonhole argument, specify the groups and explain why the counts force the conclusion.